

ADD-1(n: nonnegative integer)

```

1  a ← 0
2  for i ← 1 to n
3      a ← a + i
4  return a

```

$$\Rightarrow \sum_{i=1}^n 1 = \underbrace{1+1+1+\dots+1}_{n \text{ times}} = n \cdot 1 = n \Rightarrow \underline{\underline{O(n)}}$$

ADD-3(n: nonnegative integer)

```

1  a ← 0
2  for i ← 1 to n
3      for j ← 1 to n
4          a ← a + 1
5  return a

```

$$\Rightarrow \sum_{i=1}^n \underbrace{\sum_{j=1}^n 1}_n = \sum_{i=1}^n n = n \cdot n = n^2 \Rightarrow \underline{\underline{O(n^2)}}$$

ADD-GENERALISED(n: nonnegative integer)

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1  for i1 ← 1 to n
2      for i2 ← 1 to n
3          ...
4          for ik ← 1 to n
5              expression

```

$$\Rightarrow \sum_{i_1=1}^n \sum_{i_2=1}^n \dots \sum_{i_k=1}^n 1 = n^k \Rightarrow \underline{\underline{O(n^k)}}$$

ADD-4(n: nonnegative integer)

```

1  a ← 0
2  for i ← 1 to n
3      for j ← i to n
4          a ← a + 1
5  return a

```

$$\begin{aligned} \Rightarrow \sum_{i=1}^n \sum_{j=i}^n 1 &= \sum_{i=1}^n \left(\underbrace{\sum_{j=1}^n 1}_n - \underbrace{\sum_{j=1}^{i-1} 1}_{i-1} \right) = \sum_{i=1}^n (n - (i-1)) = \sum_{i=1}^n (n - i + 1) \\ &= \sum_{i=1}^n (n+1) - \sum_{i=1}^n i = (n \cdot (n+1)) - \left(\frac{n \cdot (n+1)}{2} \right) = \frac{n \cdot (n+1)}{2} = \frac{n^2 + n}{2} \\ &\Rightarrow \underline{\underline{O(n^2)}} \end{aligned}$$

ADD-5(n: nonnegative integer)

```

1  a ← 0
2  for i ← 1 to n
3      for j ← i+1 to n
4          a ← a + 1
5  return a

```

$$\begin{aligned} \Rightarrow \sum_{i=1}^n \sum_{j=i+1}^n 1 &= \sum_{i=1}^n \left(\underbrace{\sum_{j=1}^n 1}_n - \underbrace{\sum_{j=1}^i 1}_i \right) = \sum_{i=1}^n (n - i) = \sum_{i=1}^n n - \sum_{i=1}^n i \\ &= n^2 - \frac{n \cdot (n+1)}{2} = \frac{n^2 - n}{2} \Rightarrow \underline{\underline{O(n^2)}} \end{aligned}$$

A2(n: positive integer)

```

1  a ← 0
2  for i ← 1 to n-1
3      for j ← i+1 to n
4          for k ← 1 to j
5              a ← a + 1
6  return a

```

$$\begin{aligned} \Rightarrow \sum_{i=1}^{n-1} \sum_{j=i+1}^n \underbrace{\sum_{k=1}^j 1}_j &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n j = \sum_{i=1}^{n-1} \left(\sum_{j=1}^n j - \sum_{j=1}^i j \right) = \sum_{i=1}^{n-1} \left(\frac{n(n+1)}{2} - \frac{i(i+1)}{2} \right) \\ &= \sum_{i=1}^{n-1} \frac{n(n+1)}{2} - \sum_{i=1}^{n-1} \frac{i^2 + i}{2} = (n-1) \cdot \frac{n(n+1)}{2} - \frac{1}{2} \left(\sum_{i=1}^{n-1} i^2 + \sum_{i=1}^{n-1} i \right) \end{aligned}$$

$$= (n-1) \cdot \frac{\overbrace{n(n+1)}^{n^2+n}}{2} - \frac{1}{2} \left(\frac{\overbrace{(n-1) \cdot n \cdot (2n-1)}^{n^2-n}}{6} + \frac{(n-1) \cdot n}{2} \right) = \frac{n^3-n}{2} - \frac{1}{12} (2n^3 - 3n^2 + n) - \frac{1}{4} (n^2 - n)$$

$$= \cancel{\frac{n^3}{2}} - \frac{n}{2} - \cancel{\frac{n^3}{6}} + \cancel{\frac{n^2}{4}} - \frac{n}{12} - \cancel{\frac{n^2}{4}} + \frac{n}{4} = \frac{1}{3} n^3 - \frac{1}{3} n = \frac{1}{3} (n^3 - n) \Rightarrow \underline{\underline{O(n^3)}}$$

P1(a₁, a₂, ..., a_n: array of positive integers)

```

1 s ← 0
2 for i ← 1 to n-4
3   for j ← i to i+4
4     for k ← i to j
5       s ← s + ai

```

(15(n-4))

$$\Rightarrow \sum_{i=1}^{n-4} \sum_{j=i}^{i+4} \sum_{k=i}^j 1 = \sum_{i=1}^{n-4} \sum_{j=i}^{i+4} \left(\sum_{k=1}^j 1 - \sum_{k=1}^{i-1} 1 \right) = \sum_{i=1}^{n-4} \sum_{j=i}^{i+4} (j - i + 1)$$

$$= \sum_{i=1}^{n-4} \left(\sum_{j=i}^{i+4} j + 1 - \sum_{j=i}^{i+4} 1 \right) = \sum_{i=1}^{n-4} \left((5i + 15) - (5i) \right) = \sum_{i=1}^{n-4} 15$$

$$= 15(n-4) = 15n - 60 \Rightarrow \underline{\underline{O(n)}}$$

$$T(n) = 2T(n-1) + 1 \Rightarrow T(n) = 2 \cdot T(n-1) + 1$$

$$T(0) = 0$$

$$= 2 \cdot (2 \cdot T(n-2) + 1) + 1 = 4 \cdot T(n-2) + 3$$

$$= 2 \cdot (2 \cdot (2 \cdot T(n-3) + 1) + 1) + 1 = 8 \cdot T(n-3) + 7$$

$$= 16 \cdot T(n-4) + 15$$

$$\Rightarrow T(n) = 2^k \cdot T(n-k) + (2^k - 1), \quad k = n$$

$$\Rightarrow T(n) = 2^n \cdot \underbrace{T(0)}_0 + 2^n - 1 = 2^n - 1 \Rightarrow \underline{\underline{O(2^n)}}$$

$$T(n) = T(n-1) + n \Rightarrow T(n) = T(n-1) + n$$

$$T(0) = 1$$

$$= T(n-2) + (n-1) + n$$

$$= T(n-3) + (n-2) + (n-1) + n$$

$$\vdots$$

$$T(n) = T(n-k) + n - (k-1) + n - (k-2) + \dots + n$$

$$k=n \Rightarrow T(n) = T(0) + 1 + 2 + \dots + n = 1 + \frac{n \cdot n + 1}{2} = \frac{1}{2} n^2 + \frac{1}{2} n + 1 \Rightarrow \underline{\underline{O(n^2)}}$$

$$T(n) = 2T\left(\frac{n}{2}\right) + 1 \Rightarrow T(n) = 2T\left(\frac{n}{2}\right) + 1$$

$$= 4T\left(\frac{n}{4}\right) + 3$$

$$= 8T\left(\frac{n}{8}\right) + 7$$

$$\vdots$$

$$\Rightarrow T(n) = 2^k \cdot T\left(\frac{n}{2^k}\right) + 2^k - 1$$

$$2^k = n \Rightarrow T(n) = n \cdot T(1) + n - 1$$

$$= 2n - 1 \Rightarrow \underline{\underline{O(n)}}$$

$$F(0) = 0 \text{ and } F(1) = 1$$

$$F(n) = F(n-1) + F(n-2)$$

$$\Rightarrow x^2 = x + 1 \Rightarrow x^2 - x - 1 = 0 \Rightarrow x_1 = \frac{1+\sqrt{5}}{2}, \quad x_2 = \frac{1-\sqrt{5}}{2}$$

$$\Rightarrow F(n) = (x_1)^n + (x_2)^n \Rightarrow F(n) = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$\Rightarrow \underline{\underline{O\left(\left(\frac{1+\sqrt{5}}{2}\right)^n\right)}} \approx \underline{\underline{O(1.6180^n)}}$$

Bubble Sort: eleman sayısı = length[A] = n

for i ← 1 to length[A] (C₁)

do for j ← length[A] downto i + 1 (C₂) ⇒ T(n) = c₁ · (n+1) + c₂ · ∑_{i=1}ⁿ (n-i+1) + c₃ · ∑_{i=1}ⁿ (n-i) + c₄ · ∑_{i=1}ⁿ (n-i)

do if A[j] < A[j-1] (C₃)

(C₄) then exchange A[j] ↔ A[j-1]

$$= c_1 \cdot n + c_1 + c_2 \cdot \sum_{i=1}^n 1 + c_2 \cdot \sum_{i=1}^n (n-i) + c_3 \cdot \sum_{i=1}^n (n-i) + c_4 \cdot \sum_{i=1}^n (n-i)$$

$$= \left(c_1 \cdot n + c_2 \cdot \sum_{i=1}^n 1 \right) + \left((c_2 + c_3 + c_4) \cdot \sum_{i=1}^n (n-i) \right) + c_1 = (c_1 + c_2) \cdot n + (c_2 + c_3 + c_4) \cdot \frac{n^2 - n}{2} + c_1$$

$$\sum_{i=1}^n (n-i) = \sum_{i=1}^n n - \sum_{i=1}^n i = \frac{n(n+1)}{2} - n = \frac{n^2 + n}{2} - n = \frac{n^2 - n}{2} \Rightarrow O(n) + O(n^2) + O_1 = \underline{O(n^2)}$$

Selection Sort:

n ← length[A] (C₁)

for j ← 1 to n - 1 (C₂)

do ek ← j (C₃)

for i ← j + 1 to n (C₄)

do if A[i] < A[ek] (C₅)

(C₆) then ek ← i

(C₇) exchange A[j] ↔ A[ek]

$$\Rightarrow T(n) = c_1 + c_2 \cdot n + c_3 \cdot (n-1) + c_4 \cdot \sum_{j=1}^{n-1} (n-j+1) + c_5 \cdot \sum_{j=1}^{n-1} (n-j) + c_6 \cdot \sum_{j=1}^{n-1} (n-j) + c_7 \cdot (n-1)$$

$$= \left(c_1 + c_2 \cdot n + c_3 \cdot (n-1) + c_4 \cdot \sum_{j=1}^{n-1} 1 \right) + \left((c_4 + c_5 + c_6) \cdot \sum_{j=1}^{n-1} (n-j) \right)$$

$$= c_1 - c_3 - c_4 + \left(c_2 \cdot n + c_3 \cdot n + c_4 \cdot n \right) + \left((c_4 + c_5 + c_6) \cdot \frac{n^2 - n}{2} \right)$$

$$\sum_{j=1}^{n-1} (n-j) = (n-1) \cdot n - \frac{(n-1) \cdot n}{2} = \frac{n^2 - n}{2} \Rightarrow O(1) + O(n) + O(n^2) = \underline{O(n^2)}$$

Quick Sort:

En Kötü Durum:

$$T(n) = T(1) + T(n-1) + \alpha n$$

$$= T(1) + (T(1) + T(n-2) + \alpha(n-1)) + \alpha n = 2T(1) + T(n-2) + \alpha(n+n-1)$$

$$= 3T(1) + T(n-3) + \alpha(n + (n-1) + (n-2))$$

$$= k \cdot T(1) + T(n-k) + \alpha \cdot \sum_{i=0}^{k-1} (n-i)$$

$$= k \cdot T(1) + T(n-k) + \alpha \cdot \left(\sum_{i=0}^{k-1} n - \sum_{i=0}^{k-1} i \right)$$

$$= k \cdot T(1) + T(n-k) + \alpha \cdot \left(k \cdot n - \frac{(k-1) \cdot k}{2} \right)$$

$$= k \cdot T(1) + T(n-k) + \alpha \cdot \left(\frac{2kn - k^2 + k}{2} \right), \quad \begin{matrix} n-k=1 \\ k=(n-1) \end{matrix}$$

$$= (n-1) \cdot T(1) + T(1) + \alpha \cdot \left(\frac{2n^2 - 2n}{2(n-1)} \cdot n - \frac{n^2 - 2n + 1}{2} + (n-1) \right)$$

$$= n \cdot T(1) + \alpha \left(\frac{n^2 + n - 2}{2} \right) = O(1) + O(n) + O(n^2) = \underline{O(n^2)}$$

En İyi Durum:

$$\begin{aligned}T(n) &= 2 T\left(\frac{n}{2}\right) + \alpha n \\&= 4 T\left(\frac{n}{4}\right) + \alpha 2n \\&= 8 T\left(\frac{n}{8}\right) + \alpha 3n \\&= 2^k \cdot T\left(\frac{n}{2^k}\right) + \alpha \cdot kn, \quad \begin{matrix} n/2^k = 1 \\ 2^k = n \quad k = \log_2 n \end{matrix} \\&= n \cdot T(1) + \alpha \cdot n \cdot \log_2 n \\&= O(n) + O(n \log_2 n) = \underline{\underline{O(n \log_2 n)}}$$

Merge Sort:

En Kötü Durum:

$$\begin{aligned}T(1) &= O \\T(n) &= 2 \cdot T\left(\frac{n}{2}\right) + (n-1) \quad \text{Karşılaştırma Sayısı} \\&= 4 \cdot T\left(\frac{n}{4}\right) + (n-2) + (n-1) \\&= 8 \cdot T\left(\frac{n}{8}\right) + (n-4) + (n-2) + (n-1) \\&= 2^k \cdot T\left(\frac{n}{2^k}\right) + \sum_{i=1}^k (n - 2^{i-1}), \quad 2^k = n \\&= n \cdot T(1) + \sum_{i=1}^{\log_2 n} (n - 2^{i-1}) \\&= n \cdot 0 + \sum_{i=1}^{\log_2 n} n - \frac{1}{2} \sum_{i=1}^{\log_2 n} 2^i \\&= n \cdot \log_2 n - \frac{1}{2} \left(\frac{2^{\log_2 n + 1} - 1}{2 - 1} - 1 \right) \\&= n \cdot \log_2 n - \frac{1}{2} ((2 \cdot 2^{\log_2 n}) - 2) \\&= n \cdot \log_2 n - (2^{\log_2 n} - 1) \\&= n \cdot \log_2 n - n + 1 \Rightarrow \underline{\underline{O(n \log_2 n)}}$$

En İyi Durum:

$$\begin{aligned}T(1) &= O \\T(n) &= 2 \cdot T\left(\frac{n}{2}\right) + \frac{n}{2} \quad \text{Karşılaştırma Sayısı} \\&= 4 T\left(\frac{n}{4}\right) + \frac{n}{2} + \frac{n}{2} \\&= 8 \cdot T\left(\frac{n}{8}\right) + \frac{n}{2} + \frac{n}{2} + \frac{n}{2} \\&= 2^k \cdot T\left(\frac{n}{2^k}\right) + k \cdot \frac{n}{2}, \quad 2^k = n\end{aligned}$$

$$= n \cdot T(1) + \log_2 n \cdot \frac{n}{2}$$

$$= \log_2 n \cdot \frac{n}{2} \Rightarrow \underline{\underline{O(n \cdot \log_2 n)}}$$

Insertion Sort :

```

for j ← 2 to n (C1)
  do key ← A[j] (C2)
  (C3) ▷ A[j]'yi sıralı A[1 . . j - 1] dizisine ekle
  (C4) i ← j - 1
  (C5) while i > 0 and A[i] > key
    (C6) do A[i + 1] ← A[i]
    (C7) i ← i - 1
  (C8) A[i + 1] ← key
  
```

En Kötü Durum :

$$T(n) = c_1 \cdot \sum_{j=2}^n 1 + c_2 \cdot (n-1) +$$

$$1) T(n) = T\left(\frac{n}{3}\right) + 1$$

$$\Rightarrow \begin{aligned} a &= 1 \\ b &= \frac{3}{2} \\ f(n) &= n^0 \end{aligned} \quad \begin{aligned} f(n) &= \Theta(n^{\log_b a} \cdot \log^k n) \\ n^0 &= \Theta(n^{\log_{\frac{3}{2}} 1} \cdot \log^0 n) \Rightarrow T(n) = \Theta(\log n) \end{aligned}$$

$$2) T(n) = 16 T\left(\frac{n}{4}\right) + n$$

$$\Rightarrow \begin{aligned} a &= 16 \\ b &= 4 \\ f(n) &= n \end{aligned} \quad \begin{aligned} f(n) &= O(n^{\log_b a - \epsilon}) \\ n &= O(n^{\log_4 16 - 1}) \Rightarrow T(n) = \underline{\underline{\Theta(n^2)}} \end{aligned}$$

$$3) T(n) = 2 T\left(\frac{n}{2}\right) + n \cdot \log n$$

$$\Rightarrow \begin{aligned} a &= 2 \\ b &= 2 \\ f(n) &= n \cdot \log n \end{aligned} \quad \begin{aligned} f(n) &= \Theta(n^{\log_b a} \cdot \log^k n) \\ n \cdot \log n &= \Theta(n^{\log_2 2} \cdot \log^1 n) \Rightarrow T(n) = \underline{\underline{\Theta(n \cdot \log^2 n)}} \end{aligned}$$

$$4) T(n) = T\left(\frac{n}{2}\right) + 2^n$$

$$\Rightarrow \begin{aligned} a &= 1 \\ b &= 2 \\ f(n) &= 2^n \end{aligned} \quad \begin{aligned} f(n) &= \Omega(n^{\log_b a + \epsilon}) \\ 2^n &= \Omega(n^{\log_2 1 + \epsilon}) \\ 1. f\left(\frac{n}{2}\right) \leq c \cdot f(n) &\Rightarrow 2^{\frac{n}{2}} \leq c \cdot 2^n \end{aligned} \} \Rightarrow T(n) = \underline{\underline{\Theta(2^n)}}$$

$$5) T(n) = 64 T\left(\frac{n}{8}\right) - n^2 \log n$$

$$\Rightarrow f(n) = -n^2 \log n \Rightarrow f(n) < 0 \Rightarrow \underline{\underline{\text{Ana teorema ile çözülemez!}}}$$

$$6) T(n) = 6 T\left(\frac{n}{3}\right) + n^2 \log n$$

$$\Rightarrow \begin{aligned} a &= 6 \\ b &= 3 \\ f(n) &= n^2 \log n \end{aligned} \quad \begin{aligned} f(n) &= \Omega(n^{\log_b a + \epsilon}) \\ n^2 \log n &= \Omega(n^{\log_3 6 + \epsilon}) \\ 6 f\left(\frac{n}{3}\right) \leq c \cdot f(n) &\Rightarrow 6 \left(\frac{n^2}{9} \log \frac{n}{3}\right) \leq c \cdot n^2 \log n \end{aligned} \} \Rightarrow T(n) = \Theta(n^2 \log n)$$

$$7) T(n) = 3T\left(\frac{n}{2}\right) + n^2$$

$$\Rightarrow \begin{array}{l} a=3 \\ b=2 \\ f(n)=n^2 \end{array} \quad \begin{array}{l} f(n) = \Omega(n^{\log_2 3 + \epsilon}) \\ n^2 = \Omega(n^{\log_2 3 + \epsilon}) \\ 3 \cdot f\left(\frac{n}{2}\right) \leq c \cdot f(n) \Rightarrow 3 \cdot \frac{n^2}{4} \leq c \cdot n^2 \end{array} \left. \vphantom{\begin{array}{l} a=3 \\ b=2 \\ f(n)=n^2 \end{array}} \right\} \Rightarrow T(n) = \underline{\underline{\Theta(n^2)}}$$

$$8) T(n) = 4T\left(\frac{n}{2}\right) + n^2$$

$$\Rightarrow \begin{array}{l} a=4 \\ b=2 \\ f(n)=n^2 \end{array} \quad \begin{array}{l} f(n) = \Theta(n^{\log_2 4} \cdot \log^k n) \\ n^2 = \Theta(n^{\log_2 4} \cdot \log^0 n) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(n^2 \log n)}}$$

$$9) T(n) = \sqrt{2} T\left(\frac{n}{2}\right) + \log n$$

$$\Rightarrow \begin{array}{l} a=\sqrt{2} \\ b=2 \\ f(n)=\log n \end{array} \quad \begin{array}{l} f(n) = O(n^{\log_b a - \epsilon}) \\ \log n = O(n^{\log_2 \sqrt{2} - \epsilon}) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(n^{\log_2 \sqrt{2}})}} = \underline{\underline{\Theta(\sqrt{n})}}$$

$$10) T(n) = 0,5 \cdot T\left(\frac{n}{2}\right) + 1/n$$

$$\Rightarrow a=0,5 \Rightarrow a < 1 \Rightarrow \underline{\underline{\text{Ana teoren ile qozqilemez!}}}$$

$$11) T(n) = 2^n T\left(\frac{n}{2}\right) + n^n$$

$$\Rightarrow a=2^n \Rightarrow a \notin \mathbb{R} \Rightarrow \underline{\underline{\text{Ana teoren ile qozqilemez!}}}$$

$$12) T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$\Rightarrow \begin{array}{l} a=2 \\ b=2 \\ f(n)=n \end{array} \quad \begin{array}{l} f(n) = \Theta(n^{\log_2 2} \cdot \log^k n) \\ n = \Theta(n^{\log_2 2} \cdot \log^0 n) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(n \cdot \log n)}}$$

$$13) T\left(\frac{n}{2}\right) + \Theta(1)$$

$$\Rightarrow \begin{array}{l} a=1 \\ b=2 \\ f(n)=n^0 \end{array} \quad \begin{array}{l} f(n) = \Theta(n^{\log_2 1} \cdot \log^k n) \\ n^0 = \Theta(n^{\log_2 1} \cdot \log^0 n) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(\log n)}}$$

$$14) 2T\left(\frac{n}{4}\right) + 1$$

$$\Rightarrow \begin{matrix} a=2 \\ b=4 \\ f(n)=n^0 \end{matrix} \quad \begin{matrix} f(n) = O(n^{\log_b a - \epsilon}) \\ n^0 = O(n^{\log_4 2 - \frac{1}{2}}) \end{matrix} \Rightarrow T(n) = \Theta(n^{\log_4 2}) = \underline{\underline{\Theta(\sqrt{n})}}$$

$$15) 2T\left(\frac{n}{4}\right) + \sqrt{n}$$

$$\Rightarrow \begin{matrix} a=2 \\ b=4 \\ f(n)=\sqrt{n} \end{matrix} \quad \begin{matrix} f(n) = \Theta(n^{\log_b a} \cdot \log^k n) \\ \sqrt{n} = \Theta(n^{\log_4 2} \cdot \log^0 n) \end{matrix} \Rightarrow T(n) = \underline{\underline{\Theta(\sqrt{n} \cdot \log n)}}$$

$$16) 2T\left(\frac{n}{4}\right) + n$$

$$\Rightarrow \begin{matrix} a=2 \\ b=4 \\ f(n)=n \end{matrix} \quad \begin{matrix} f(n) = \Omega(n^{\log_b a + \epsilon}) \\ n = \Omega(n^{\log_4 2 + \frac{1}{2}}) \end{matrix} \quad \left. \begin{matrix} 2f\left(\frac{n}{4}\right) \leq c \cdot f(n) \Rightarrow 2\left(\frac{n}{4}\right) \leq c \cdot n \end{matrix} \right\} \Rightarrow T(n) = \underline{\underline{\Theta(n)}}$$

$$17) 2T\left(\frac{n}{4}\right) + n^2$$

$$\Rightarrow \begin{matrix} a=2 \\ b=4 \\ f(n)=n^2 \end{matrix} \quad \begin{matrix} f(n) = \Omega(n^{\log_b a + \epsilon}) \\ n^2 = \Omega(n^{\log_4 2 + \frac{3}{2}}) \end{matrix} \quad \left. \begin{matrix} 2f\left(\frac{n}{4}\right) \leq c \cdot f(n) \Rightarrow 2 \cdot \frac{n^2}{4} \leq c \cdot n^2 \end{matrix} \right\} \Rightarrow T(n) = \underline{\underline{\Theta(n^2)}}$$

$$18) 2T\left(\frac{n}{2}\right) + n^4$$

$$\Rightarrow \begin{matrix} a=2 \\ b=2 \\ f(n)=n^4 \end{matrix} \quad \begin{matrix} f(n) = \Omega(n^{\log_b a + \epsilon}) \\ n^4 = \Omega(n^{\log_2 2 + 3}) \end{matrix} \quad \left. \begin{matrix} 2f\left(\frac{n}{2}\right) \leq c \cdot f(n) \Rightarrow 2 \cdot \left(\frac{n^4}{16}\right) \leq c \cdot n^4 \end{matrix} \right\} \Rightarrow T(n) = \underline{\underline{\Theta(n^4)}}$$

$$19) T(n) = T\left(\frac{7n}{10}\right) + n$$

$$\Rightarrow \begin{matrix} a=1 \\ b=10/7 \\ f(n)=n \end{matrix} \quad \begin{matrix} f(n) = \Omega(n^{\log_b a + \epsilon}) \\ n = \Omega(n^{\log_{10/7} 1 + 1}) \end{matrix} \quad \left. \begin{matrix} 1 \cdot f\left(\frac{7n}{10}\right) \leq c \cdot f(n) \Rightarrow \frac{7n}{10} \leq c \cdot n \end{matrix} \right\} \Rightarrow T(n) = \underline{\underline{\Theta(n)}}$$

$$20) T(n) = 16 \cdot T\left(\frac{n}{4}\right) + n^2$$

$$\Rightarrow \begin{array}{l} a=16 \\ b=4 \\ f(n)=n^2 \end{array} \quad \begin{array}{l} f(n) = \Theta(n^{\log_b a} \cdot \log^k n) \\ n^2 = \Theta(n^{\log_4 16} \cdot \log^0 n) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(n^2 \log n)}}$$

$$21) T(n) = 7 T\left(\frac{n}{3}\right) + n^2$$

$$\Rightarrow \begin{array}{l} a=7 \\ b=3 \\ f(n)=n^2 \end{array} \quad \begin{array}{l} f(n) = \Omega(n^{\log_b a + \epsilon}) \\ n^2 = \Omega(n^{\log_3 7 + \epsilon}) \end{array} \quad \left. \begin{array}{l} 7 \cdot f\left(\frac{n}{3}\right) \leq c \cdot f(n) \Rightarrow 7 \cdot \frac{n^2}{9} \leq c \cdot n^2 \end{array} \right\} \Rightarrow \underline{\underline{\Theta(n^2)}}$$

$$22) T(n) = 7 \cdot T\left(\frac{n}{2}\right) + n^2$$

$$\Rightarrow \begin{array}{l} a=7 \\ b=2 \\ f(n)=n^2 \end{array} \quad \begin{array}{l} f(n) = O(n^{\log_b a - \epsilon}) \\ n^2 = O(n^{\log_2 7 - \epsilon}) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(n^{\log_2 7})}}$$

$$23) 2 T\left(\frac{n}{4}\right) + \sqrt{n}$$

$$\Rightarrow \begin{array}{l} a=2 \\ b=4 \\ f(n)=\sqrt{n} \end{array} \quad \begin{array}{l} f(n) = \Theta(n^{\log_b a} \cdot \log^k n) \\ \sqrt{n} = \Theta(n^{\log_4 2} \cdot \log^0 n) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(\sqrt{n} \cdot \log n)}}$$

$$24) T(n) = 3 T\left(\frac{n}{4}\right) + n \log n$$

$$\Rightarrow \begin{array}{l} a=3 \\ b=4 \\ f(n)=n \log n \end{array} \quad \begin{array}{l} f(n) = \Omega(n^{\log_b a + \epsilon}) \\ n \log n = \Omega(n^{\log_4 3 + \epsilon}) \end{array} \quad \left. \begin{array}{l} 3 \cdot f\left(\frac{n}{4}\right) \leq c \cdot f(n) \Rightarrow 3 \cdot \frac{n}{4} \cdot \log \frac{n}{4} \leq c \cdot n \log n \end{array} \right\} \Rightarrow \underline{\underline{T(n) = \Theta(n \log n)}}$$

$$25) T(n) = 2 \cdot T\left(\frac{n}{2}\right) + n \log n$$

$$\Rightarrow \begin{array}{l} a=2 \\ b=2 \\ f(n)=n \log n \end{array} \quad \begin{array}{l} f(n) = \Theta(n^{\log_b a} \cdot \log^k n) \\ n \log n = \Theta(n^{\log_2 2} \cdot \log^1 n) \end{array} \Rightarrow T(n) = \underline{\underline{\Theta(n \log^2 n)}}$$

LCS (Longest Common Subsequence) Algoritması:

1) "CREM" ve "RACECAR" ikiye adet katar olmak üzere bu iki katarın en uzun ortak alt katarı nedir?

⇒

	O	R	A	C	E	C	A	R
O	0	0	0	0	0	0	0	0
C	0	0	0	1	1	1	1	1
R	0	1	1	1	1	1	1	2
E	0	1	1	1	2	2	2	2
A	0	1	2	2	2	2	3	3
M	0	1	2	2	2	2	3	3

⇒ "REA" veya "CEA"

2) "bacad" ve "accbadcb" ikiye adet katar olmak üzere bu iki katarın en uzun ortak alt katarı nedir?

⇒

	O	a	c	c	b	a	d	c	b
O	0	0	0	0	0	0	0	0	0
b	0	0	0	0	1	1	1	1	1
a	0	1	1	1	1	2	2	2	2
c	0	1	2	2	2	2	2	3	3
a	0	1	2	2	2	3	3	3	3
d	0	1	2	2	2	3	4	4	4

⇒ "acad"

3) "xyzzyx" ve "zxygzxz" ikiye adet katar olmak üzere bu iki katarın en uzun ortak alt katarı nedir?

⇒

	O	z	x	y	y	z	x	z
O	0	0	0	0	0	0	0	0
x	0	0	1	1	1	1	1	1
z	0	1	1	1	1	2	2	2
y	0	1	1	2	2	2	2	2
z	0	1	1	2	2	3	3	3
y	0	1	1	2	3	3	3	3
x	0	1	2	2	3	3	4	4

⇒ "zyyx" veya "xyzz"

4) "bdcaba" ve "abcadbab" ikiye adet katar olmak üzere bu iki katarın en uzun ortak alt katarı nedir?

⇒

	O	a	b	c	b	d	a	b
O	0	0	0	0	0	0	0	0
b	0	0	1	1	1	1	1	1
d	0	0	1	1	1	2	2	2
c	0	0	1	2	2	2	2	2
a	0	1	1	2	2	2	3	3
b	0	1	2	2	3	3	3	4
a	0	1	2	2	3	3	4	4

⇒ "bcba"

A 10x10 grid showing the addition of two numbers in base 10. The grid contains digits from 0 to 9. Red arrows indicate the path of the addition, starting from the bottom-left (9,9) and moving towards the top-right (0,0). Blue arrows indicate the carry path, starting from the bottom-left and moving towards the top-right. The grid is labeled with 'x' and 'y' axes.

$$\Rightarrow \underline{\underline{'zxyzyx'}}$$

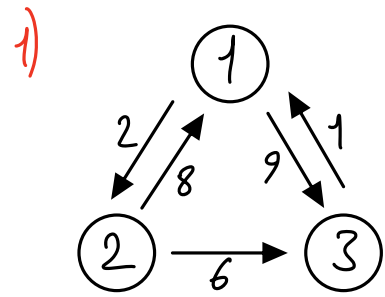
A 10x10 grid with two circles at (1,9) and (2,10), and two arrows pointing to them.

LCS $\Rightarrow$ $O(n \cdot n) \rightarrow O(m \cdot n)$
 $O(2^{m+n}) \Rightarrow O(\max(m, n))$

Floyd-Warsh $\Rightarrow O(n^3) \rightarrow O(n^2)$
 Shortest Paths $\Rightarrow O(n.w) \rightarrow O(n.w)$ } Q11
 $O(2^n)$
 $O(n \log n) \Rightarrow O(n)$ } Kserin

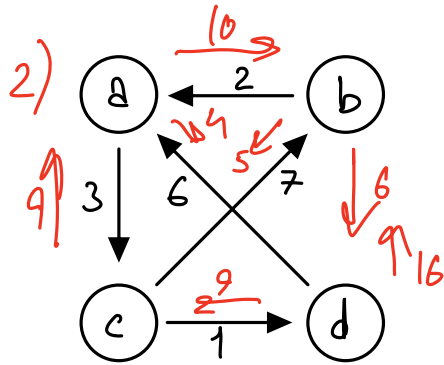
[illegible]

Floyd - Warshall Algoritması :



Düğümlerin arasındaki en kısa mesafeleri bul.

$$\Rightarrow D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix} \Rightarrow D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix} \Rightarrow D_2 = \begin{bmatrix} 0 & 2 & 8 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix} \Rightarrow D_3 = \underline{\underline{\begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}}}$$



Düğümlerin arasındaki en kısa mesafeleri bul.

$$\Rightarrow D_0 = \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & \infty \\ \infty & 7 & 0 & 1 \\ 6 & \infty & \infty & 0 \end{bmatrix} \Rightarrow D_a = \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & 5 & \infty \\ \infty & 7 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix} \Rightarrow D_b = \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & 5 & \infty \\ 9 & 7 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix}$$

$$D_c = \begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & 6 \\ 9 & 7 & 0 & 1 \\ 6 & 16 & 9 & 0 \end{bmatrix} \Rightarrow D_d = \underline{\underline{\begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & 6 \\ 7 & 7 & 0 & 1 \\ 6 & 16 & 9 & 0 \end{bmatrix}}}$$

0/1 Sırt Çantası Algoritması:

1) $((2,3), (3,4), (4,5), (5,6))$ şeklinde ağırlıkları ve değerleri verilen 4 eleman kapasitesi 5 birim olan bir çantaya hangileri konulursa çantadaki elemanların değeri en fazla olur?

⇒

	0	1	2	3	4	5
0	0	0	0	0	0	0
2,3	1	0	0	3	3	3
3,4	2	0	0	3	4	4
4,5	3	0	0	3	4	5
5,6	4	0	0	3	4	5

⇒ 1. ve 2. eleman

2) $((5,10), (4,10), (6,30), (3,50))$ şeklinde ağırlıkları ve değerleri verilen 4 eleman kapasitesi 10 birim olan bir çantaya hangileri konulursa çantadaki elemanların değeri en fazla olur?

⇒

	0	1	2	3	4	5	6	7	8	9	10
0	0	0	0	0	0	0	0	0	0	0	0
5,10	1	0	0	0	0	10	10	10	10	10	10
4,10	2	0	0	0	0	10	10	10	10	50	50
6,30	3	0	0	0	0	10	10	10	10	50	70
3,50	4	0	0	0	50	50	50	90	90	90	90

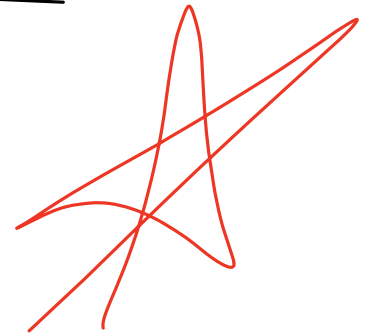
⇒ 2. ve 4. eleman

3) $((5,30), (10,20), (20,100), (30,90), (40,160))$ şeklinde ağırlıkları ve değerleri verilen 5 eleman kapasitesi 60 birim olan bir çantaya hangileri konulursa çantadaki elemanların değeri en fazla olur?

⇒

	0	5	10	15	20	25	30	35	40	45	50	55	60
0	0	0	0	0	0	0	0	0	0	0	0	0	0
5,30	1	0	30	30	30	30	30	30	30	30	30	30	30
10,20	2	0	30	30	50	50	50	50	50	50	50	50	50
20,100	3	0	30	30	50	100	130	130	130	130	130	130	130
30,90	4	0	30	30	50	100	130	130	150	150	150	190	220
40,160	5	0	30	30	50	100	130	130	150	160	190	220	280

⇒ 3. ve 5. eleman



4) $(3, 100), (2, 190), (2, 160)$ şeklinde ağırlıkları ve değerleri verilen 3 eleman kapasitesi 4 birim olan bir çantaya hangileri konulursa çantadaki elemanların değeri en fazla olur?

⇒

	0	1	2	3	4	5
0						
1						
2						
3						
4						

